Effects of Multiple Sclerosis Relapse Treatment During Pregnancy on Offspring Neurocognitive Development and Behaviour at School Age

V. Kozik[a]*, M. Dreiling[a]*, S. Thiel[b], K. Hellwig[b], D. Müller[a], O. Tiedge[c], M. Schwab[a], F. Rakers[a]

[a]University Hospital Jena, Germany [b]Ruhr University Bochum, Germany [c]NeuroZentrum Erfurt, Germany

MS relapse during pregnancy → MS guideline → **High-dose glucocorticoid** (GC) treatment, considered safe for foetus

Clinical data

Adverse effect of prenatal exposure to the GC betamethasone (2×12mg) to improve fetal lung maturation in threatened premature on neurocognitive abilities[1].

! Dose of GC in MS-relapse treatment is up to **50 x higher** than in this obstetric indication

Lower IQ and increased ADHD symptoms

? How does the developmental effect of **prenatal methylprednisolone (MP) exposure** for MS relapse treatment manifest in **school-age neurocognition**?

Method

Children of mothers with MS

cross-section: 2 groups	non-MP-exposed	n = 30 + 30	MP-exposed
age, M [SD]	10.6y [1.5]		10.1y [1.7]
female sex	40%		37%
parents higher education	60%		67%
dose MP, median	-		5g

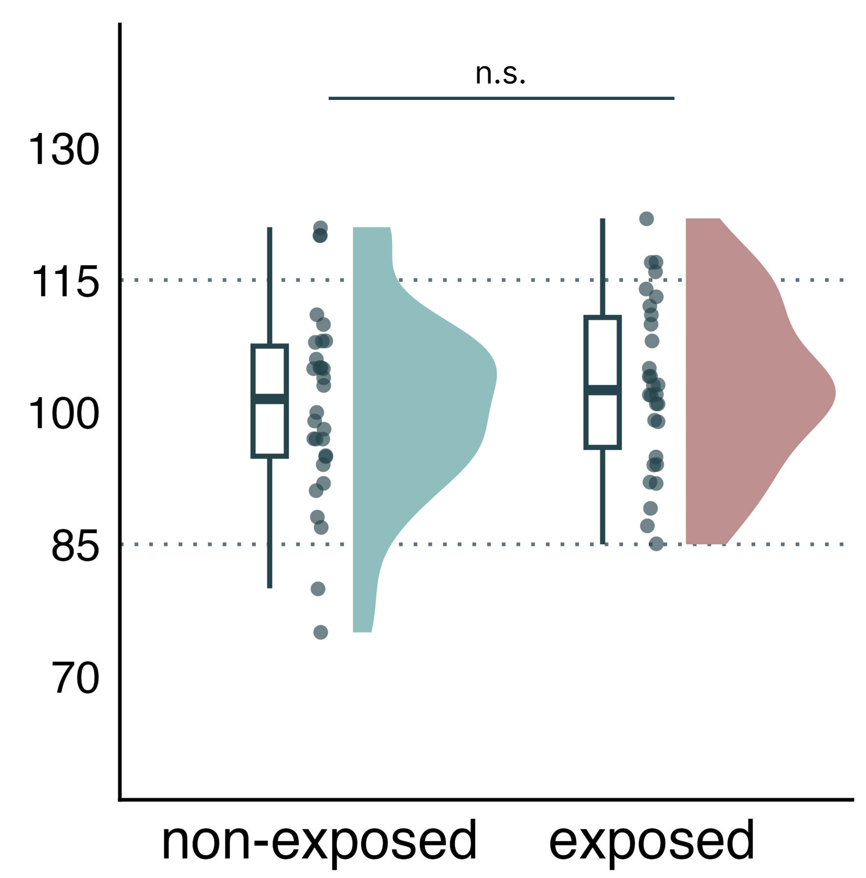
Neuropsychological assessment[2]

Intelligence	(Reynolds Intellectual Assessment Scales)
Attention	(Continuous Performance Test)
Emotional excitability	(child → standard questionnaire)
ADHD symptoms	(parent → standard questionnaire)
Behavioural difficulties	(parent → standard questionnaires)
Motor development	(Movement Assessment Battery for Children)

Analysis: robust regression modelling using MM-estimation and Huber psi weighting function, controlling for sex and parental education

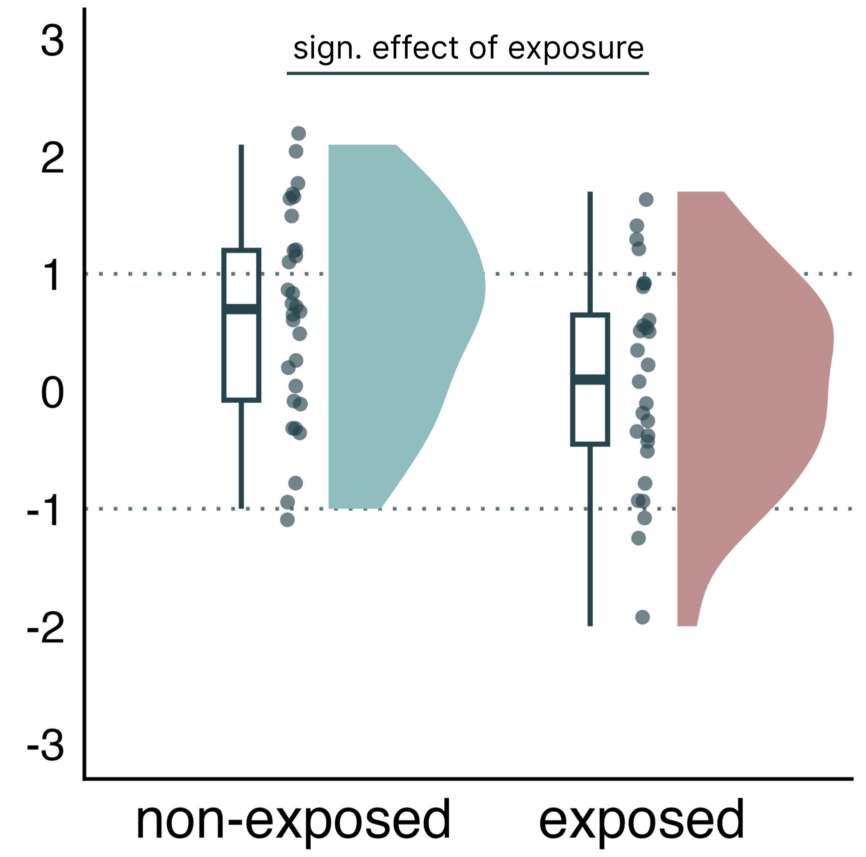
Results

Intelligence*



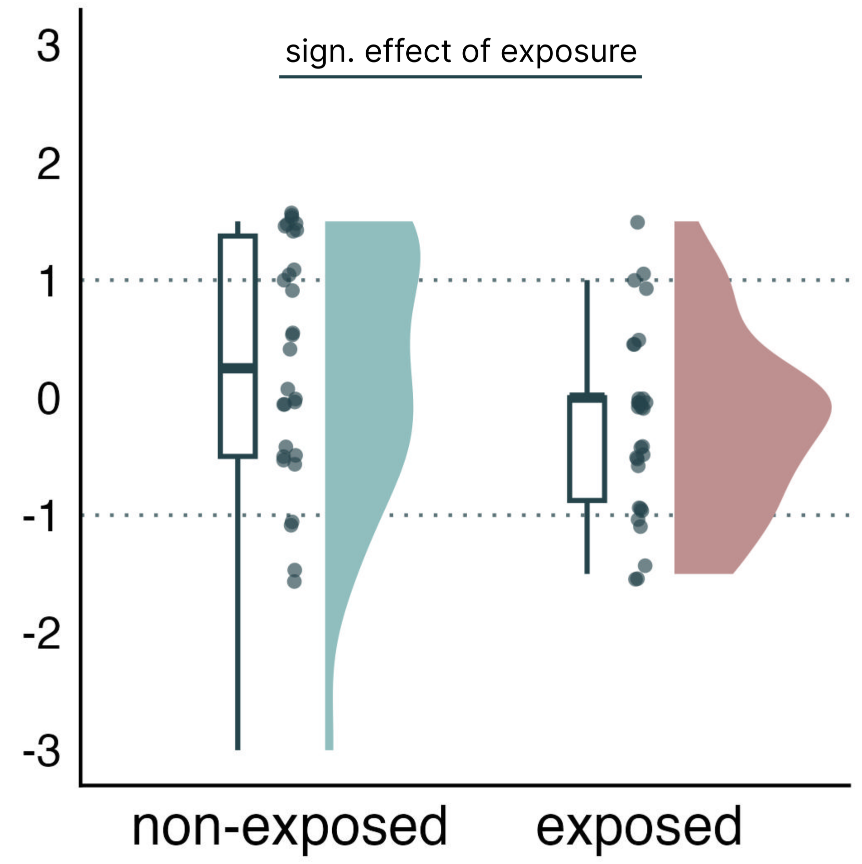
*IQ scale, RIAS

Emotional excitability*



*z-scores

CPT emission errors*



*z-scores (recoded)

No observable effect on any other cognitive or behavioural measures

No observable effect of dose or pregnancy week of exposure

Conclusions

- We observed **no long-term effect** of MS relapse therapy with MP during pregnancy on **child IQ** at 10 years of age.
- We observed some evidence for **differences in behaviour and emotional processing** compared to controls.
- Limitations: retrospective study design and relatively small sample size

Use of MP for MS relapse treatment during pregnancy should be approached with caution until a more comprehensive database becomes available.

[1]Rakers *et al.*, *Acta Obstet Gynecol Scand* 101 (2022) [2]Study protocol: Kozik *et al.*, *Front Neurol* 13 (2022)

This work was funded by the "Grant for Multiple Sclerosis Innovation (GMSI)"